

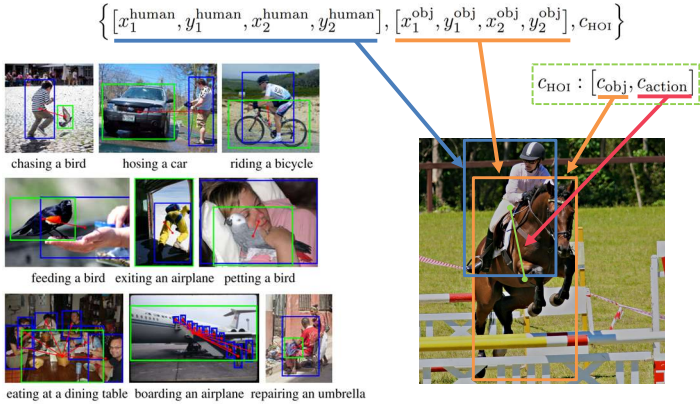
HOI-R1: Exploring the Potential of Multimodal Large Language Models for Human-Object Interaction Detection



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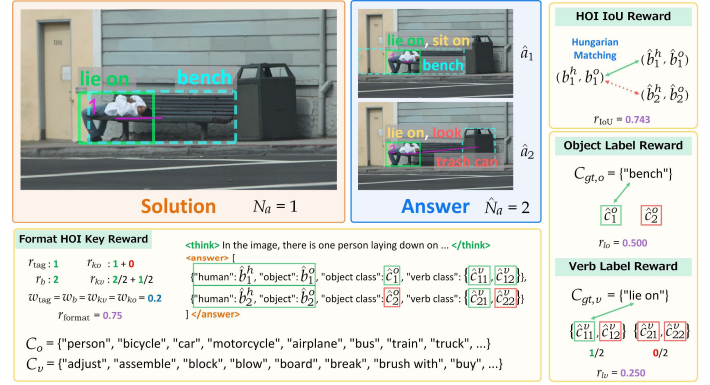


Human-Object Interaction Detection (HOID)



- In recent years, Human-Object Interaction (HOI) detection has attracted increasing attention due to its broad potential applications.
- HICO-DET, containing **47,776 images** (38,118 for training and 9,658 for testing), is one of the most widely used datasets for HOI detection.
 - It contains **600 HOI classes**, composed of **117 verb classes** and **80 object classes**

HOI-R1 Rewards



- Example Reward Functions in HOI-R1: Rewards for **HOI format, labels, and IoU** ensure the structural, semantic, and geometric consistency of the output.

$$r_{\text{format}} = w_{\text{tag}} r_{\text{tag}} + \frac{\sum_{i=1}^{N_a} \alpha_i [w_b r_{b_i} + w_{ko} r_{ko_i} + w_{kv} r_{kv_i}]}{\max(N_a, \hat{N}_a)}$$

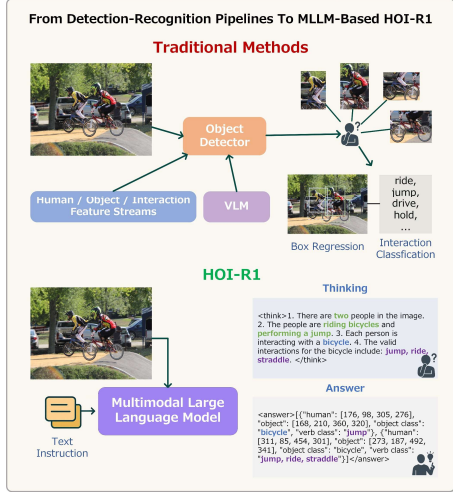
$$r_{\text{IoU}} = \frac{1}{N_a} \sum_{(i,j) \in \mathcal{M}^*} s_{ij}$$

$$s_{ij} = \frac{1}{2} [\text{IoU}(\hat{b}_i^o, \hat{b}_j^o) + \text{IoU}(\hat{b}_i^o, \hat{b}_j^o)]$$

$$r_{\text{IoU}} = \frac{\sum_{i=1}^{N_a} \alpha_i \sum_{j=1}^{|\hat{C}_{gt,v}^{(i-1)}|} 1[\hat{C}_{ij}^v \in \hat{C}_{gt,v}^{(i-1)}]}{\max(N_a, \hat{N}_a)}$$

$$r = r_{\text{format}} + r_{\text{IoU}} + r_{lv} + r_{\text{IoU}}$$

Motivation



Comparison between Conventional HODD and HOI-R1 Framework

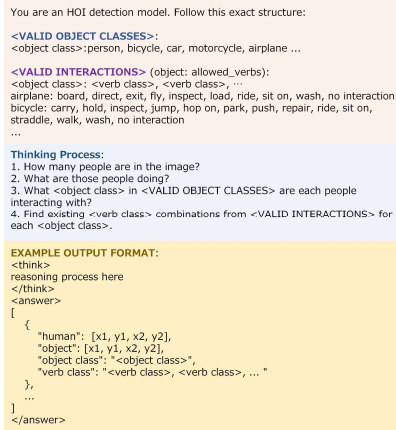
- Conventional methods **rely on object detectors**
 - Incorporating prior knowledge from VLMs requires **complex module design**
- HOI-R1 directly interprets HOIs **in natural language using an MLLM**

Results

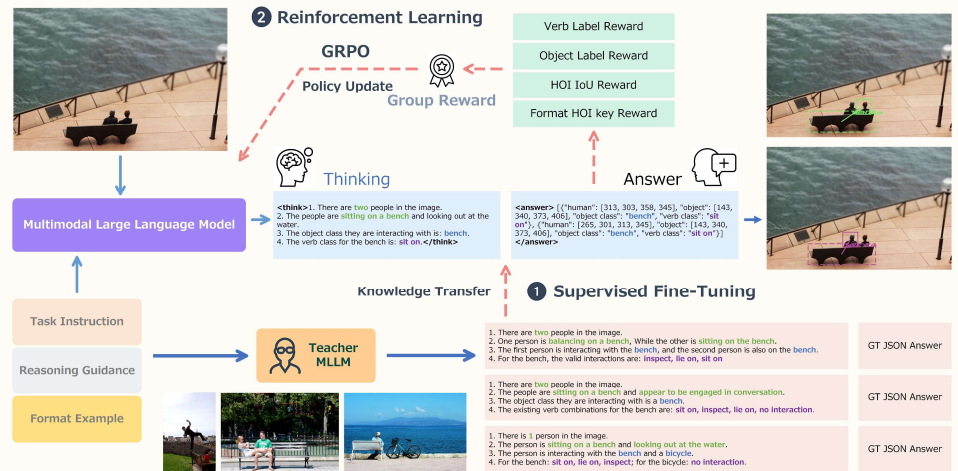
Method	Training	Default			Known Obj.		
		Full	Rare	N-Rare	Full	Rare	N-Rare
— Traditional HOID Methods —							
HO-RCNN[5]	150k	7.81	5.37	8.54	10.41	8.94	10.85
HOTR[10]	100ep	25.10	17.34	27.42	-	-	-
HOI-Trans[20]	150ep	26.61	19.15	28.84	29.13	20.98	31.57
QPC[18]	150ep	29.07	21.85	31.23	31.68	24.14	33.93
— MLLM Zero-Shot Baselines —							
Qwen2.5-VL-3B	-	8.39	9.60	8.03	8.96	9.83	8.70
Qwen2.5-VL-7B	-	10.46	14.30	9.31	11.01	14.63	9.93
Qwen2.5-VL-3B-AWQ	-	18.12	24.56	16.20	19.90	25.77	18.15
Qwen2.5-VL-72B-AWQ	-	20.71	29.62	18.05	22.93	31.87	20.26
Qwen3-VL-4B	-	16.16	22.31	14.32	17.55	22.87	15.96
Qwen3-VL-8B	-	17.41	21.96	16.05	19.17	23.39	17.91
Qwen3.5-4B	-	17.17	16.08	20.82	19.60	18.53	23.16
— SFT or RL Fine-Tuning —							
Qwen2.5-VL-3B-SFT	400	11.71	10.52	12.07	12.60	10.70	13.17
Qwen2.5-VL-3B-SFT	1ep	16.77	14.20	17.53	18.05	14.45	19.13
Qwen2.5-VL-3B-GRPO	400	12.22	14.56	11.52	13.12	14.84	12.60
Qwen2.5-VL-3B-GRPO	1ep	14.65	15.90	14.28	15.48	16.14	15.28
Qwen3-VL-4B-SFT	1ep	22.46	21.53	22.73	24.04	21.97	24.66
Rex-Omni-3B[9] (HOI-SFT)	1ep	25.30	21.24	26.51	26.97	21.79	28.51
Qwen3.5-4B-SFT	1ep	29.36	29.03	30.47	31.22	31.31	30.92
— HOI-R1 (SFT+GRPO) —							
HOI-R1-Qwen2.5-VL-3B	1ep+40	18.33	16.03	19.02	19.83	16.25	20.90
HOI-R1-Qwen3-VL-4B	1ep+40	23.35	23.58	23.28	25.12	24.11	25.42
HOI-R1-Rex-Omni-3B	1ep+40	25.24	22.57	26.04	27.00	23.21	28.14
HOI-R1-Qwen3.5-4B	1ep+40	29.79	30.39	29.60	31.61	30.67	31.89

Comparison Results on HICO-DET (mAP)

HOI-R1 Framework



HOI-R1 Instruction Template



- HOI-R1 instruction is composed of **task instructions**, **reasoning guidance**, and an **output-format example**
- SFT is performed using reasoning examples generated by a teacher MLLM

$$\mathcal{L}_{\text{SFT}} = -\mathbb{E}_{(x,q,r,a) \sim \mathcal{D}} \left[\sum_{t=1}^{T_r} \log \pi_{\theta}(r_t | x, q, r_{<t}) + \sum_{t=1}^{T_a} \log \pi_{\theta}(a_t | x, q, r, a_{<t}) \right]$$

- Update the Policy Model with GRPO Using Four Types of Reward Signals

$$\mathcal{J}_{\text{GRPO}} = -\mathbb{E}_{(x,q) \sim \mathcal{D}, (a_i) \sim \pi_{\theta}, (a_j) \sim \pi_{\theta}'} \left[\frac{1}{G} \sum_{i=1}^G \left[\min[s_1 \cdot \hat{A}_i, s_2 \cdot \hat{A}_i] - \beta \mathbb{D}_{\text{KL}}[\pi_{\theta} || \pi_{\theta}'] \right] \right]$$

- More Than 2 × Performance Improvement with Just 1 Epoch of SFT and 40 RL Steps

